

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Justification

Form mismatch is moderate. Both benchmark and deployment are text-only in Latin script, so there is no modality or script-direction violation [regional context infrastructure_notes.deployment_modality; writing_systems]. Tokenization (BPE, vocab 50,368, max sequence 1,024) [Q128, Q129] is suitable for typical citizen-correspondence lengths (40–400 words covers many but not all messages [regional context infrastructure_notes.expected_input_length, Q47]). However, signal-distribution mismatch persists at the form level: task inputs are filtered to remove non-Luxembourgish text via OpenLID and constrained to 40–400 words [Q47], which removes precisely the multilingual and length-extreme inputs the deployment will encounter. The benchmark's text register (formal news, dictionary examples, transcribed political speeches) [Q74, Q16, Q30] differs from informal, orthographically variable citizen messages. IF is MODERATE priority; the form-level alignment is partial because text-text matching is achieved, but distributional filtering and register narrowness reduce external validity.

Strengths

- Text-only modality matches deployment exactly [regional context infrastructure_notes.deployment_modality]
- Latin-script writing system aligns with all three administrative languages [regional context writing_systems]
- BPE tokenizer trained on broad LTZ pre-training corpus including comments, webchat, podcasts may handle orthographic variation reasonably [Q74, Q128]
- Sequence length cap of 1,024 tokens is sufficient for most short-to-medium correspondence

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both source and target are text. Length filter (40–400 words) [Q47] excludes very short and very long messages that are realistic in administrative correspondence (e.g., one-line inquiries, multi-page complaints).

IF-2: Yes — Luxembourg has 99.0% internet penetration [WEB-7] and advanced digital infrastructure [regional context infrastructure_notes.connectivity]; data-capture specifications match.

ID	Evidence
WEB-7	99.0% internet penetration — confirms text-output channel is universally accessible

IF-3: Use-case-specific differences: filtering of non-Luxembourgish text via OpenLID [Q47] removes code-switched inputs that the deployment must actually process. Pre-training data is broader [Q74] but downstream task data is narrower in register.

IF-4: External-validity concerns from (a) length filtering excluding tail of distribution, (b) language-detection filtering removing multilingual inputs, (c) register narrowness in downstream task data [Q47, Q112].

Information Gaps

- Empirical length distribution of expected citizen correspondence
- Empirical proportion of code-switched messages in real digital channels

Requires Expert Verification

- Sample-level review of incoming citizen messages to validate that benchmark form constraints (length, language-purity filtering) do not exclude operationally important inputs

Remediation

Gap 1: Length filter (40–400 words) and language-purity filter (OpenLID) exclude tail-distribution inputs the deployment will face

Recommendation: Construct a supplementary evaluation slice without length and language-purity filters to characterize model behavior on very short, very long, and multilingual inputs; report performance separately on these slices.